Ahsanul Kabir

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Education

Grand Valley State University, Grand Rapids, MI — May 2027
MS in Applied Computer Science, CGPA: 4.00/4.00

- Badge in Software Engineering

Bangladesh University of Engineering and Technology, Dhaka, Bangladesh — Feb 2021
BS in Computer Science and Engineering

- Class Representative - Class of 2021

Professional Experience

Graduate Research Assistant, *GVSU College of Computing*, Grand Rapids, MI — Aug 2025 – Aug 2026

- Deployed a machine learning–based plant disease detection platform using Google Cloud Compute Engine, Flask, Nuxt, Cloudflare, and gcloud CLI.
- Integrated GCS Bucket and PostgreSQL to support image uploads, model execution, and result archiving.
- Implemented a Flutter-based mobile application for offline park navigation with prompt engineering.
- Conducted field visits to gather user feedback, translated findings into concrete platform improvements, and communicated results to the project team.

Senior Engineer, *bKash Limited*, Dhaka, Bangladesh — Feb 2023 – Aug 2025

- Contributed to bKash's Payment Gateway, processing over 410K e-money transactions daily using Java, Python, AWS Cognito, AWS Lambda, DynamoDB, and CloudWatch.
- Redesigned payment gateway modules from serverless Lambda functions into an object-oriented microservice architecture, reducing API response time by 50%. Used Spring WebFlux, Redis, and API Gateway.
- Enhanced system security by implementing a cipher for REST APIs and configuring AWS WAF rules.
- Migrated the Merchant Notification system to a reactive architecture, enabling it to handle 7,000 requests per minute while reducing EC2, DynamoDB, and S3 costs.
- Collaborated with cross-functional teams to deliver high-quality products, actively participating in daily stand-ups, backlog refinement, and retrospectives.
- Introduced Test-Driven-development within the team for change requests.
- Built CI/CD pipelines with GitHub Actions to run automated tests on pull requests and deploy changes to test environments.
- Automated local environment setup using Terraform and AWS CLI, reducing manual configuration errors.
- Analyzed application metrics and logs using CloudWatch Logs Insights to monitor system health and support product and business decisions.

Software Engineer, *Reve Systems*, Dhaka, Bangladesh — Jan 2022 – Feb 2023
Junior Software Engineer — Jul 2021 – Dec 2021

- Developed administrative and customer-facing portals for a telecommunications company using Spring Boot, Struts, and MySQL.
- Implemented BalanceManager using Redis Pub/Sub for call and SMS authorization, enabling the transition from a postpaid to a more cost-effective prepaid business model.
- Automated the call-rate entry process by developing an email-driven solution, eliminating manual data-entry errors and reducing associated operational costs.
- Integrated Elasticsearch and Kibana for efficient log monitoring and analysis.
- Maintained documentation for clients, testers, and the product team, translating technical details into clear, accessible language.

Technical Skills

Languages	Python, Java, C#, C, C++, SQL, Bash, JavaScript
Frameworks & Libraries	Spring Boot, Flask, .NET, Struts, Node.js, Nuxt.js
Databases	DynamoDB, Redis, MySQL, PostgreSQL, MongoDB
AWS	EC2, SNS, S3, Lambda, Cognito, WAF, API Gateway, CloudWatch, Aurora
GCP	Compute Engine, Cloud SQL, Cloud Storage, VPC Firewall Rules, gcloud CLI
DevOps	Docker, Kubernetes, Jenkins, Terraform, Nginx, CI/CD
Tools	HTML, CSS, Git, MATLAB, DVC, Postman, JIRA, Maven, GraphQL, LaTeX Figma, Visual Studio, PowerShell, Locust
Monitoring	New Relic, ELK Stack, JDK Mission Control, JMeter

Projects

Edge Forestry, *GVSU College of Computing*

An AI-powered system that analyzes imagery to detect early signs of plant infections by identifying subtle variations in foliage color and canopy texture, enabling faster and more accurate treatment decisions. Deployed the platform on GCP, and currently working to incorporate satellite imagery analysis for broader forest health monitoring.

Edge Forestry Mobile, *GVSU College of Computing*

An offline-first Flutter app for iOS and Android that runs tree disease detection on-device and navigates field crews to flagged trees over offline vector maps, with no cellular coverage required. Built the local database and bidirectional sync layer that keeps field scans, exports, and imagery consistent with the web platform.

Tokenized Checkout Payment, *bKash Limited*

A fintech solution that helps merchants accept payments easily while giving bKash users a simple, convenient way to pay online through websites or mobile apps. Implemented merchant-facing transactional APIs and customer-facing payment validation APIs.

Instant Payment Notification, *bKash Limited*

A real-time notification system that keeps bKash merchants instantly updated on transactions. Designed and implemented the complete module in AWS SNS and reactive Spring WebFlux.

Airtel Advantage, *Reve Systems*

A platform that makes it easy for global telecom carriers to partner with Airtel, offering services like international voice, SMS, and DID, with reporting and call-rate APIs to streamline operations. Implemented call-rate and reporting related APIs.

Accolades

SentinelOne ThreatOps Challenges, GrrCON Cyber Security Summit 2025 – 2026

- Sep 2026: **Second Place** in Purple AI Challenge and **Second Place** in AI SIEM Challenge
- Oct 2025: **First Place** in Purple AI Challenge and **Third Place** in AI SIEM Challenge

Dean's List - Fall 2025 Mar 2026

College of Computing, Grand Valley State University

Semi Finalist Dec 2025

GR Deep Tech Pitch Competition, Grand Rapids

Star Award, H2 - 2023 May 2024

Product & Technology Division, bKash Limited